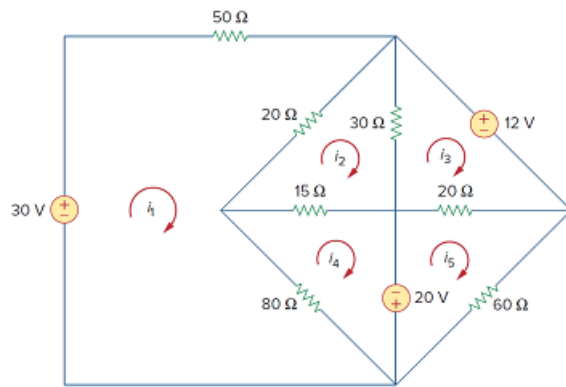


Problem

By inspection, obtain the mesh-current equations for the circuit in Fig. 3.30.

Figure 3.30:



Step-by-step solution

Step 1 of 5

Draw the circuit diagram as shown in Figure 1.

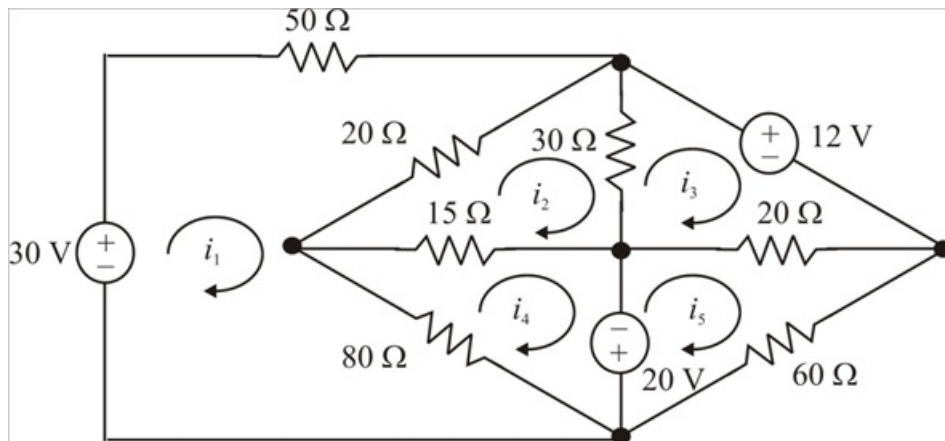


Figure 1

Step 2 of 5

If the circuit has N meshes, the mesh-current equations can be expressed in terms of resistances as:

$$\begin{pmatrix} R_{11} & \cdots & R_{1N} \\ \vdots & \ddots & \vdots \\ R_{N1} & \cdots & R_{NN} \end{pmatrix} \begin{pmatrix} i_1 \\ \vdots \\ i_N \end{pmatrix} = \begin{pmatrix} v_1 \\ \vdots \\ v_N \end{pmatrix}$$

or simply $\mathbf{R}\mathbf{i} = \mathbf{v}$

Where

R_{kk} = Sum of the resistances in mesh k

$R_{kj} = R_{jk}$ = Negative of the sum of the resistances in common with meshes k and j , $k \neq j$ $i_k =$

Unknown mesh current for mesh k in the clockwise direction

v_k = Sum taken clockwise of all independent voltage sources in mesh k , with voltage rise treated as positive

Step 3 of 5

The circuit in Figure 1 has five meshes, so we need five mesh equations. This implies that the size of the resistance matrix $\mathbf{R}$, is 5 by 5.

Each of the diagonal terms is the sum of the resistances in the related mesh, while each of the off-diagonal terms is the negative of the resistances common to meshes.

The diagonal terms of $\mathbf{R}$, in ohms can be calculated as:

$$\begin{aligned} R_{11} &= 50 + 20 + 80 \\ &= 150 \end{aligned}$$

$$\begin{aligned} R_{22} &= 20 + 30 + 15 \\ &= 65 \end{aligned}$$

$$\begin{aligned} R_{33} &= 30 + 20 \\ &= 50 \end{aligned}$$

$$\begin{aligned} R_{44} &= 80 + 15 \\ &= 95 \end{aligned}$$

$$\begin{aligned} R_{55} &= 20 + 60 \\ &= 80 \end{aligned}$$

Step 4 of 5

The off-diagonal terms in the first row are:

$$R_{12} = -20$$

$$R_{13} = 0$$

$$R_{14} = -80$$

$$R_{15} = 0$$

The off-diagonal terms in the second row are:

$$R_{21} = -20$$

$$R_{23} = -30$$

$$R_{24} = -15$$

$$R_{25} = 0$$

The off-diagonal terms in the third row are:

$$R_{31} = 0$$

$$R_{32} = -30$$

$$R_{34} = 0$$

$$R_{35} = -20$$

Step 5 of 5

The off-diagonal terms in the fourth row are:

$$R_{41} = -80$$

$$R_{42} = -15$$

$$R_{43} = 0$$

$$R_{45} = 0$$

The off-diagonal terms in the fifth row are:

$$R_{51} = 0$$

$$R_{52} = 0$$

$$R_{53} = -20$$

$$R_{54} = 0$$

The input voltage vector $\mathbf{v}$ has the following terms, in volts:

$$v_1 = 30$$

$$v_2 = 0$$

$$v_3 = -12$$

$$v_4 = 20$$

$$v_5 = -20$$

Therefore the mesh-current equations are

$$\begin{pmatrix} 150 & -20 & 0 & -80 & 0 \\ -20 & 65 & -30 & -15 & 0 \\ 0 & -30 & 50 & 0 & -20 \\ -80 & -15 & 0 & 95 & 0 \\ 0 & 0 & -20 & 0 & 80 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{pmatrix} = \begin{pmatrix} 30 \\ 0 \\ -12 \\ 20 \\ -20 \end{pmatrix}.$$